Computer Science & IT



Data Structure & programming

Graph & Hashing

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic	Heap
Topic	Build Heap
Topic	Counting
Topic	
Topic	

31-32 marks

Slide

Topics to be Covered



Topic

Topic

Topic

Topic

Topic

Graph

Graph Representation

BFS traversal

Slide



Topic : Graph

Set of vertex & edges $G(V,E)$

Real life Application

1. Network — Computer Network
   Rail Network
   pipiping system
2. Shortest path
3. Molecular structure
4. Economics
5. Map

Graph as modeling tool

Slide



Topic : Graph

Graph Representation

1 Adjacency Matrix

2 Adjacency List

Slide



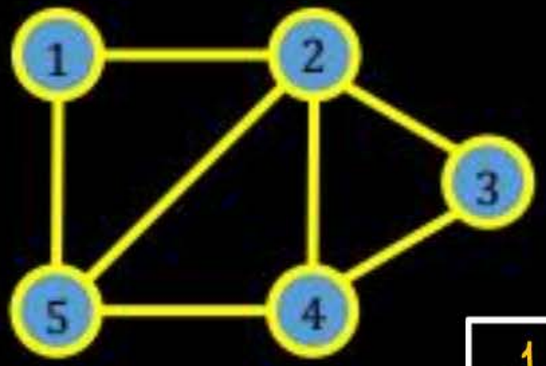
Topic : Graph Representation

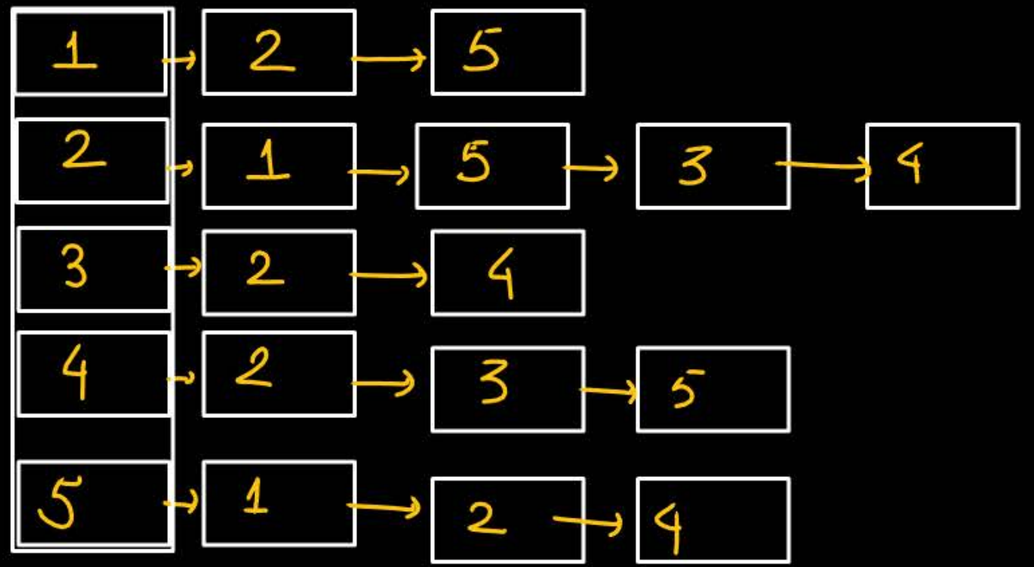
- Adjacency list
- Adjacency Matrix



Topic : Graph

Adjacency List





Simple graph : No multiple edges between vertices

No self loop

Adjacent : Direct edge

Space :

$(V+2E) : \Theta(V+E)$

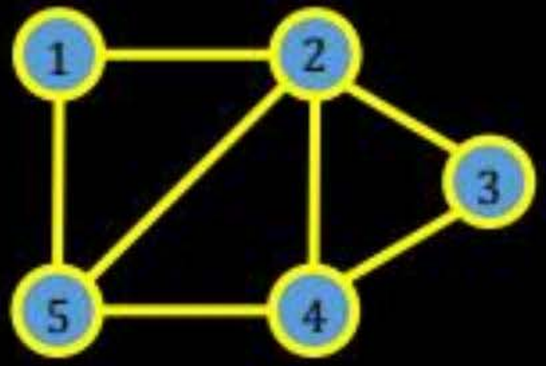
Space complexity

Length of each list is equal to degree of Node

total No. of = Sum of degree = $2 \times |E|$

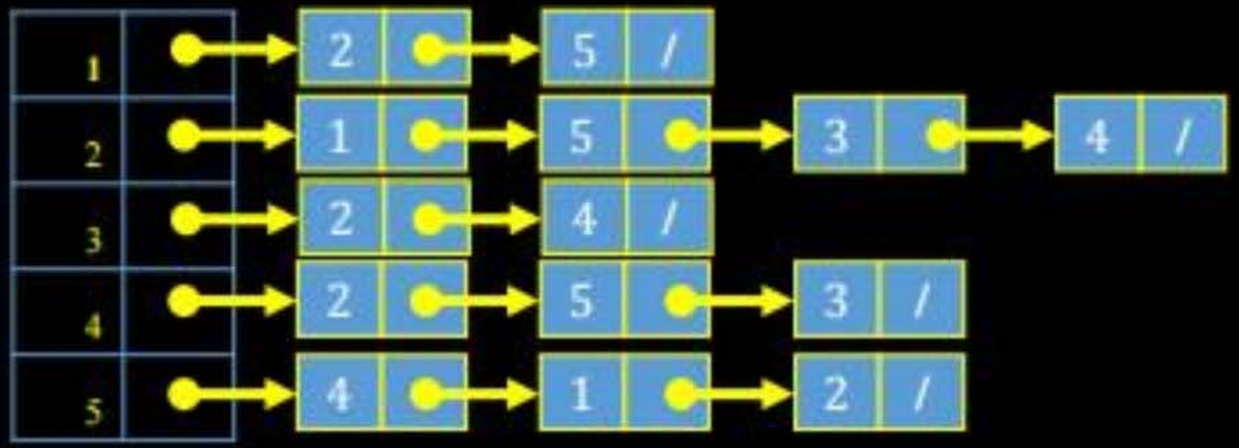


Topic : Graph



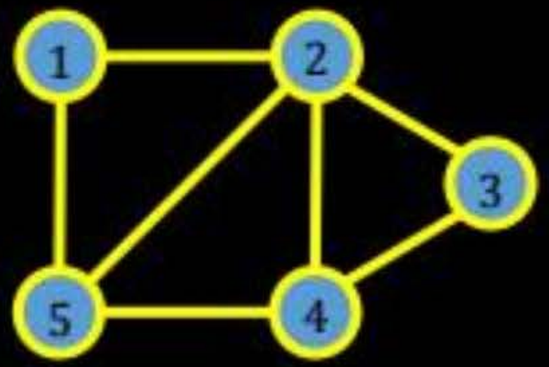
Graph is sparse graph

Adjacency List





Topic : Adjacency Matrix



M =

	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0

M[i,j] = 1 if $(i,j) \in E$

otherwis

M[i,j] = 0

Space = $(|V|)^2$

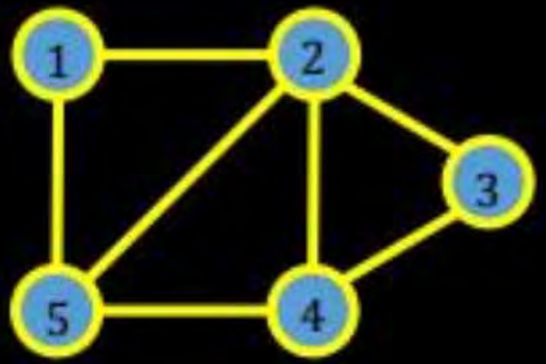
Symmetoc matoix

$M = M^T$

toanspose Matoix

Adjacency Matrix

- $a_{ij} \begin{cases} 1 & if\ (i,j) \in E \\ 0 & otherwise \end{cases}$



	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0



Topic : Graph

Graph Traversal

(1) Breadth first Search

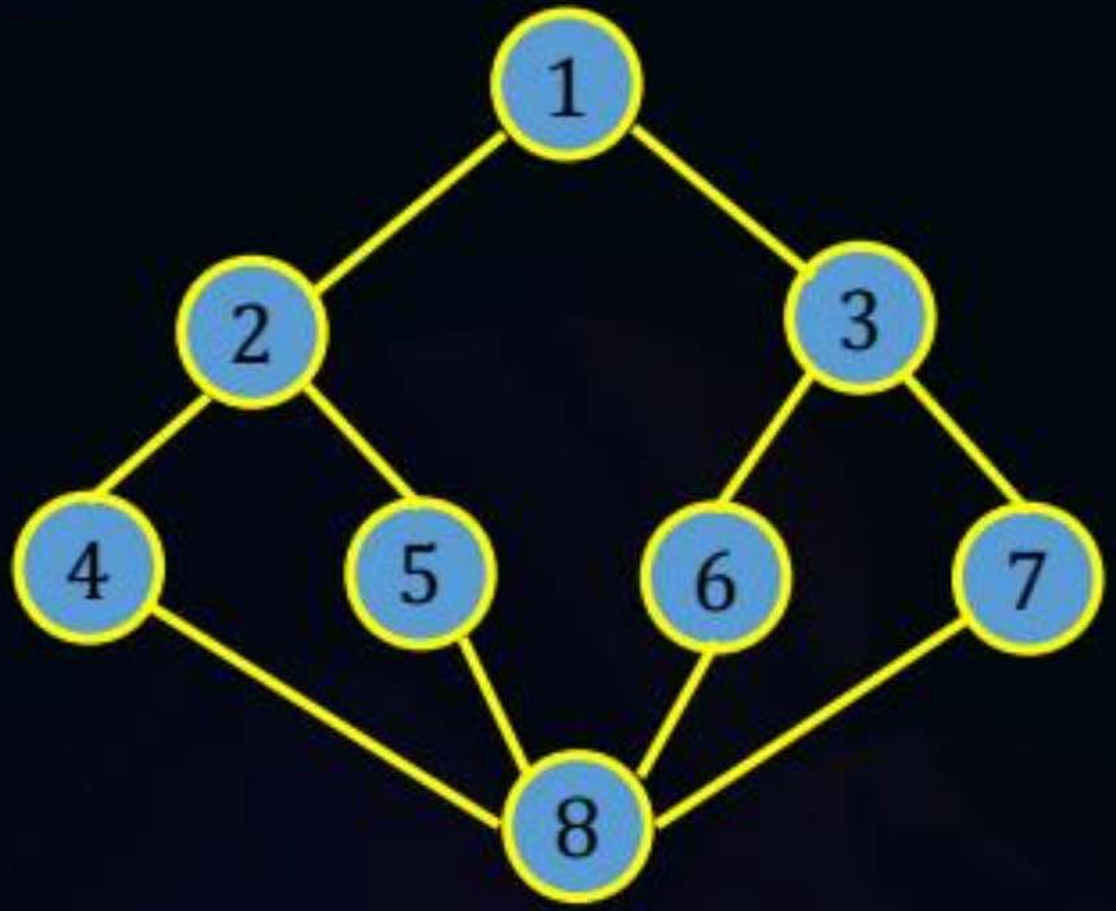
(2) Depth first Search

Traversal is process of visiting every Node of graph

Slide



Topic : Graph



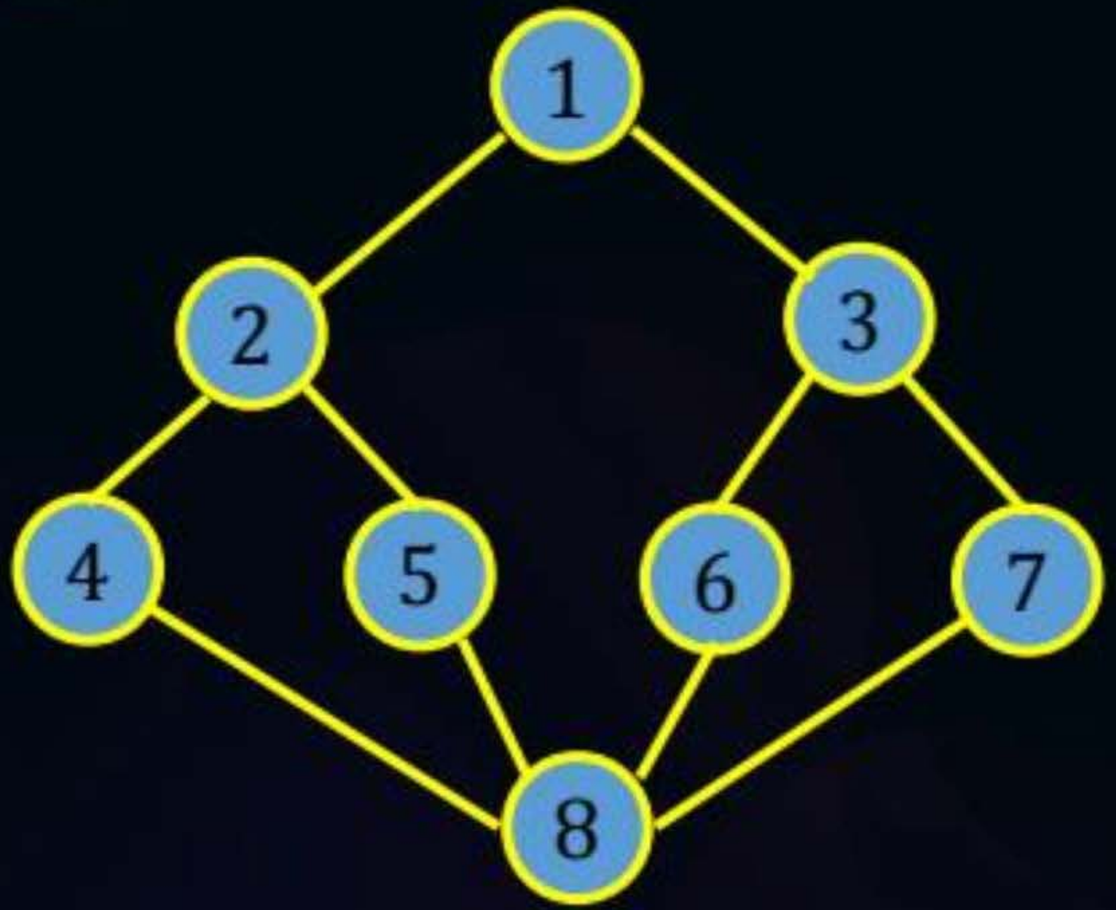
After visiting a Node the Adjacent Node of graph will be visited next.

We can start from any vertex

To maintain the order we will use queue data structure

Slide

Topic : Graph



1

Queue

3
2

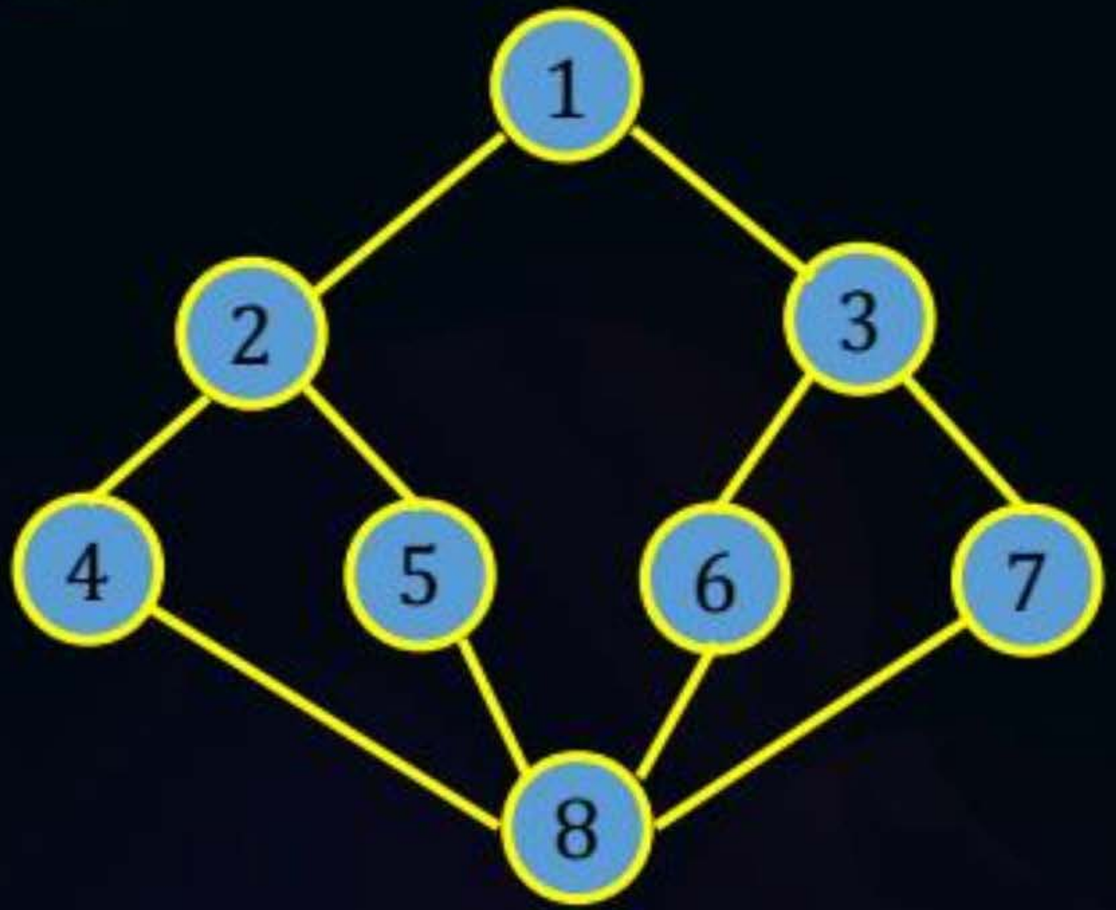
Visit the source vertex mark it visited and Enqueue Adjacent Node in queue if Not visited

2 if queue is empty then stop otherwise dequeue an element and repeat the process

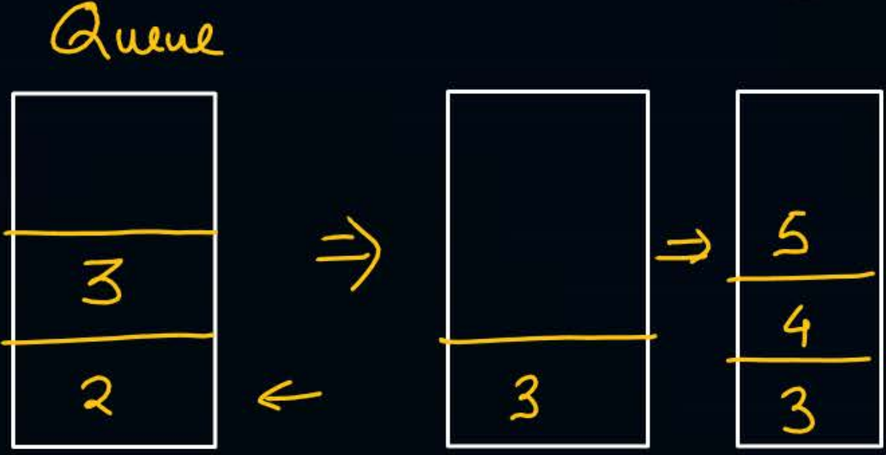
Slide



Topic : Graph

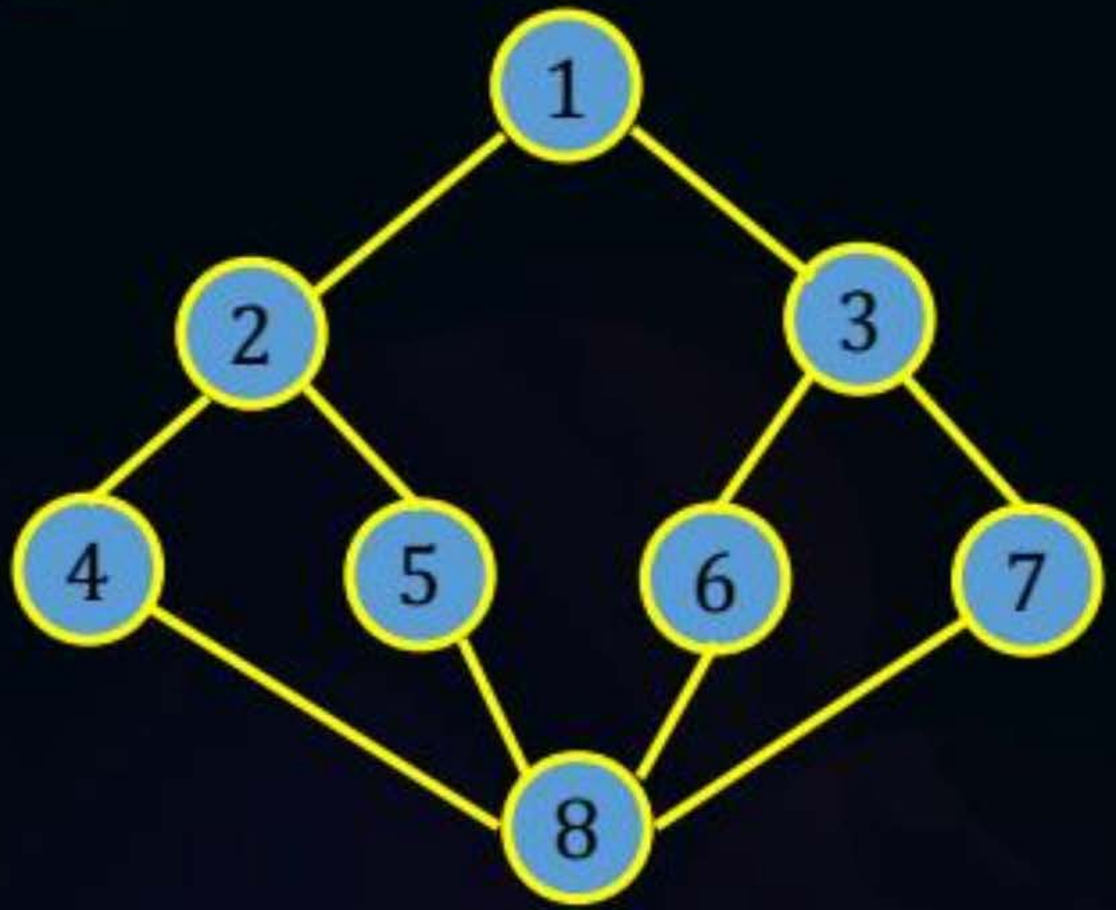


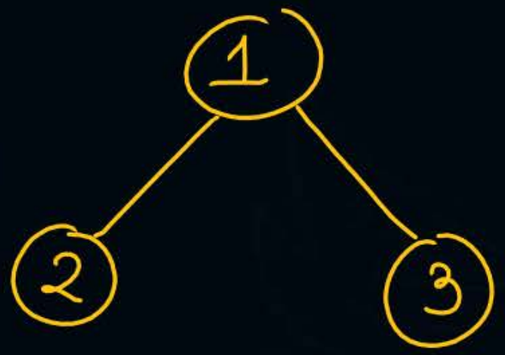




Slide

Topic : Graph





Queue



⇒



⇒



dequeue

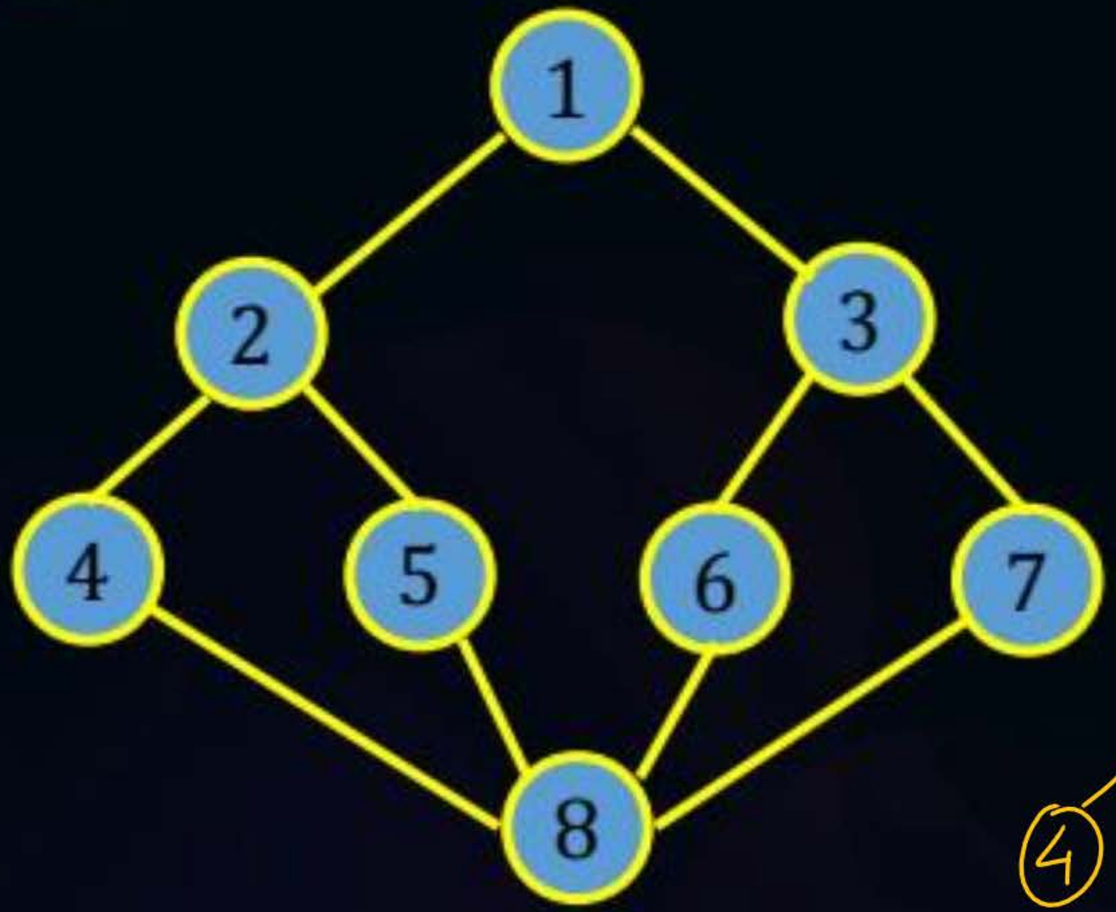
↑

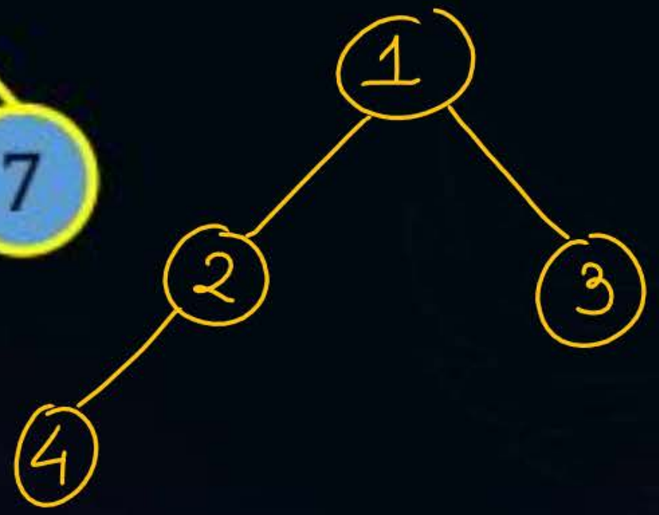
Enqueue

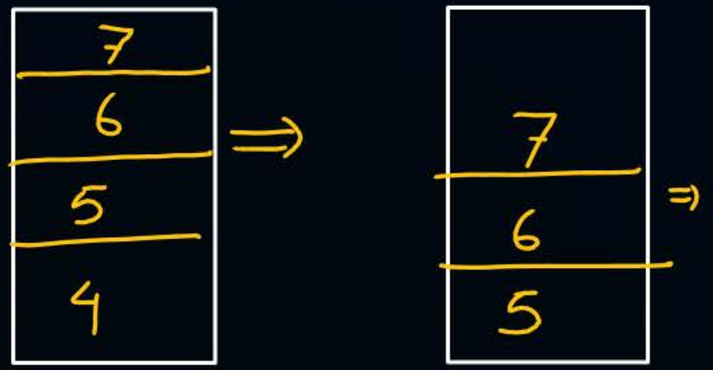
Adjacent Node of 3

Slide

Topic : Graph







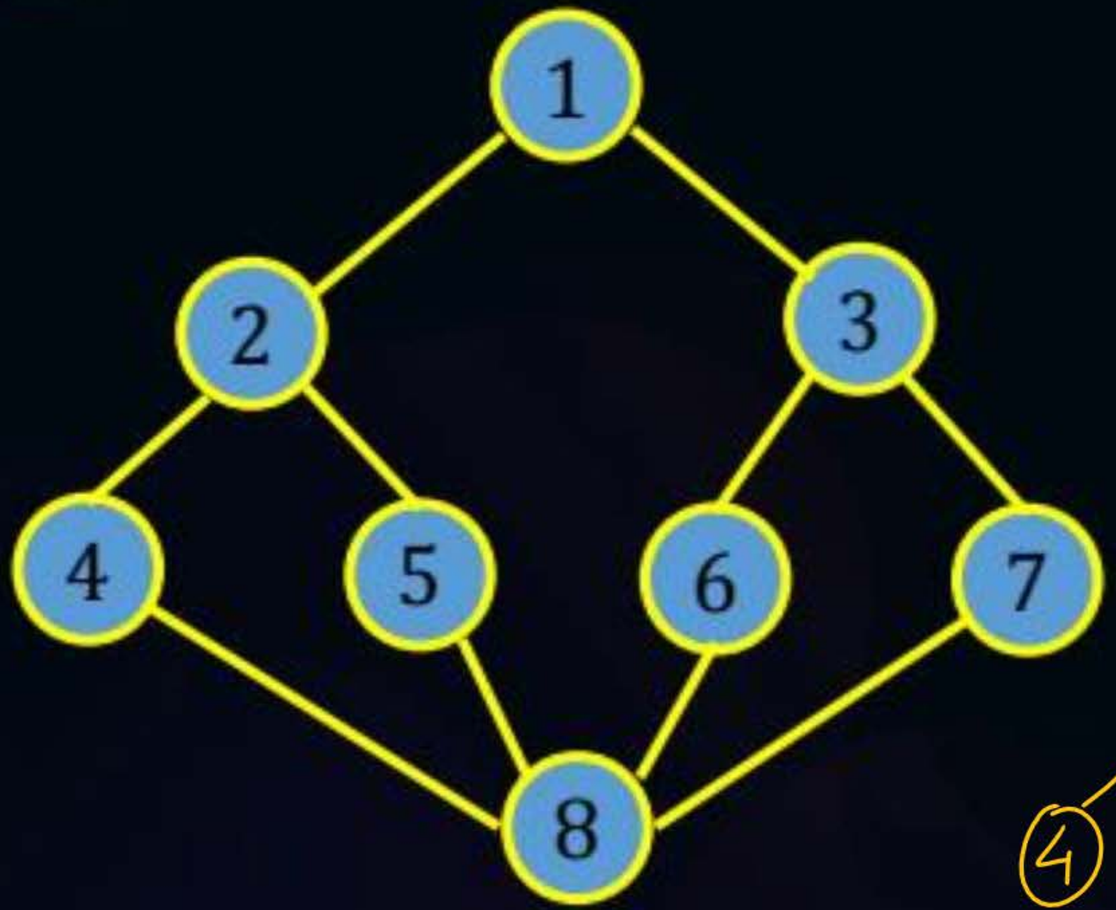
dequeue

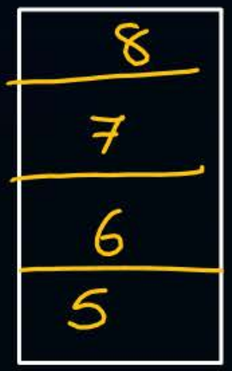
Enqueue only 8

Since 2 is already Visited

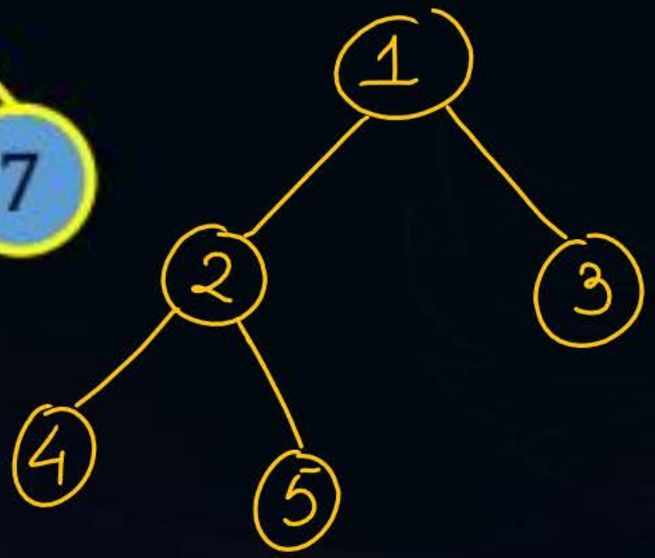
Slide

Topic : Graph





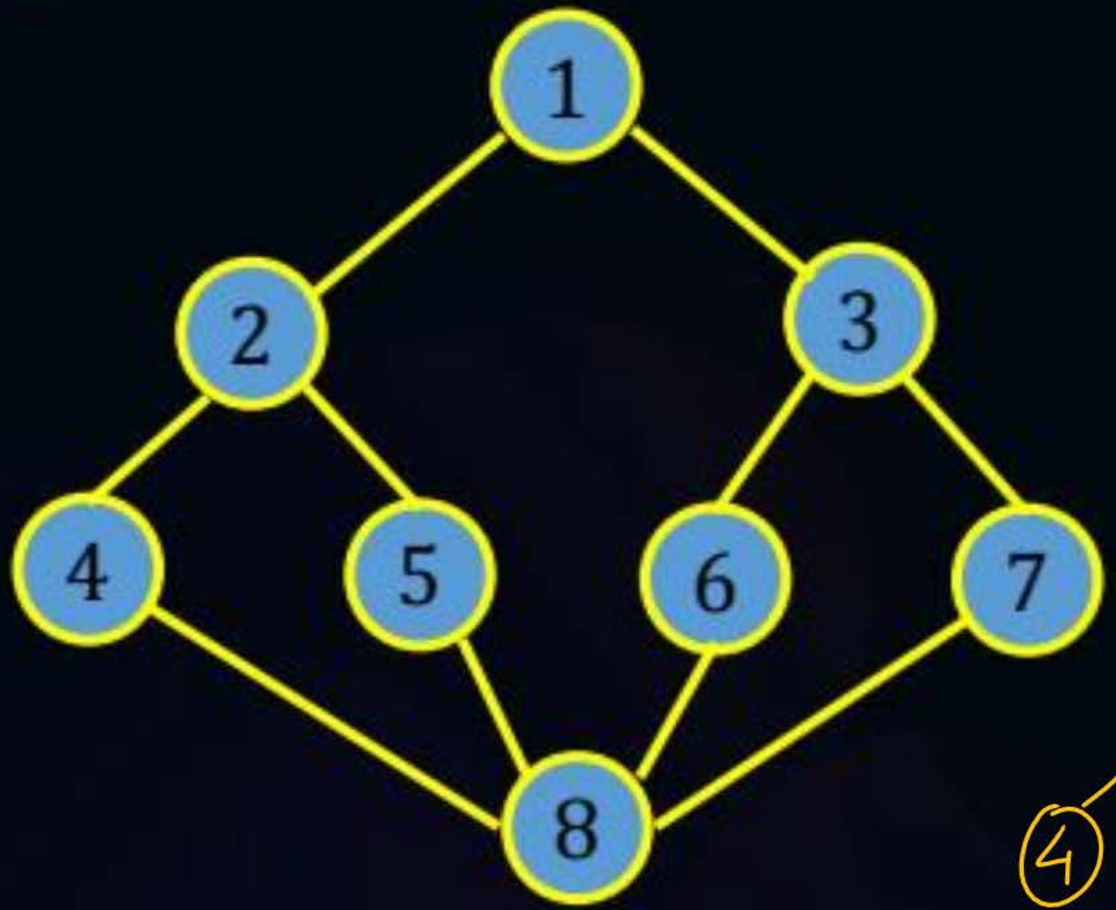


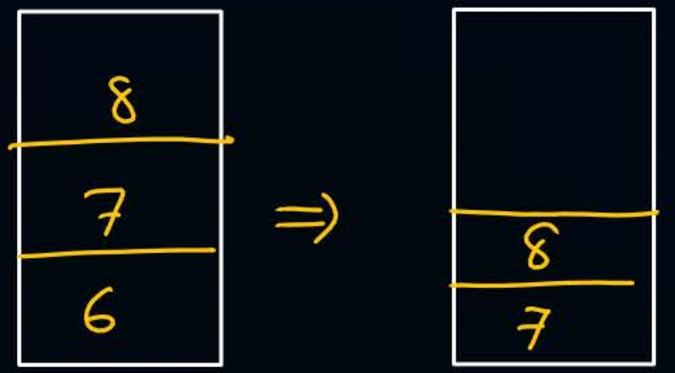


No Enqueue

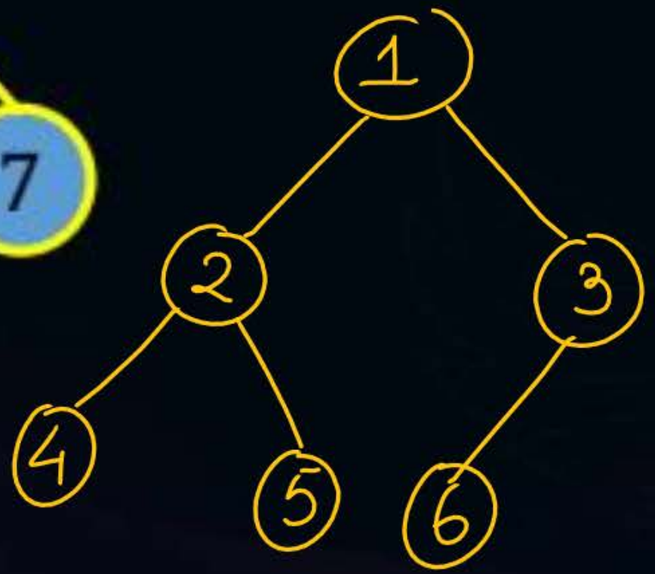
Slide

Topic : Graph





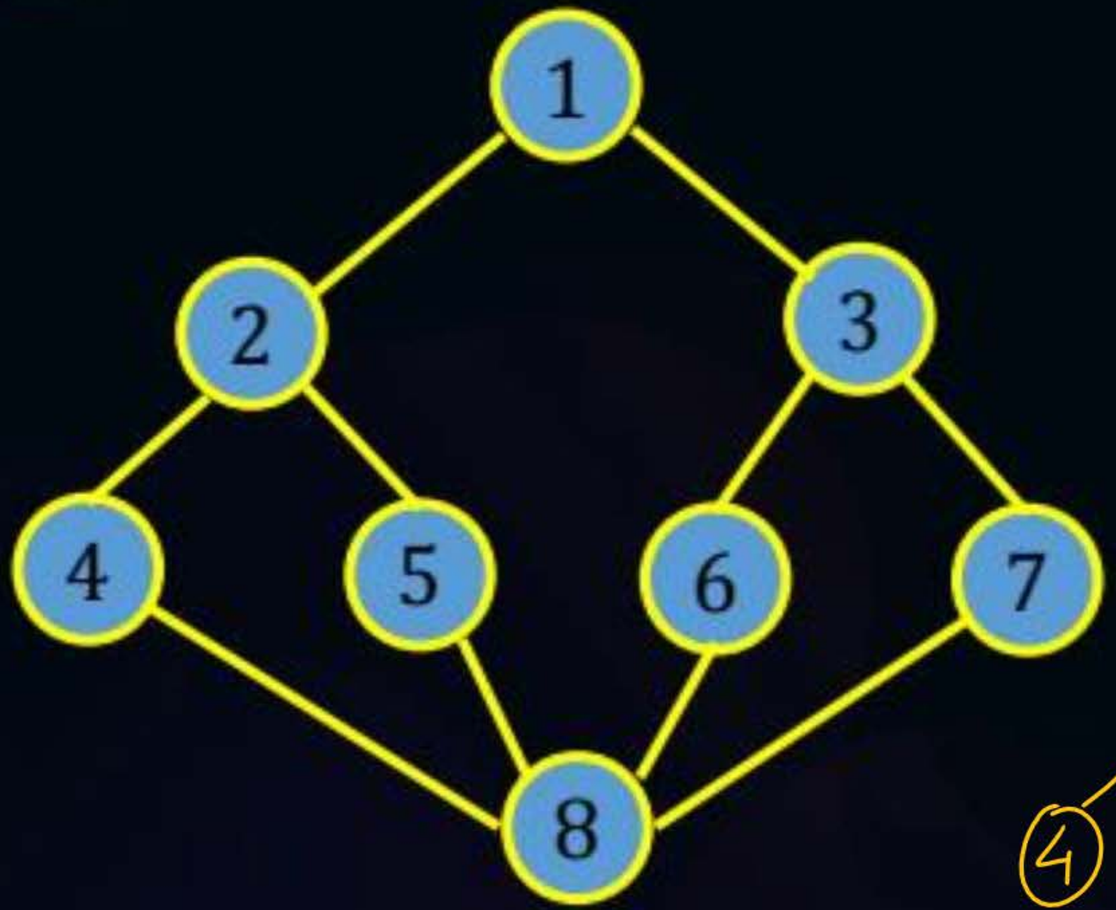
dequeue (6)



Slide



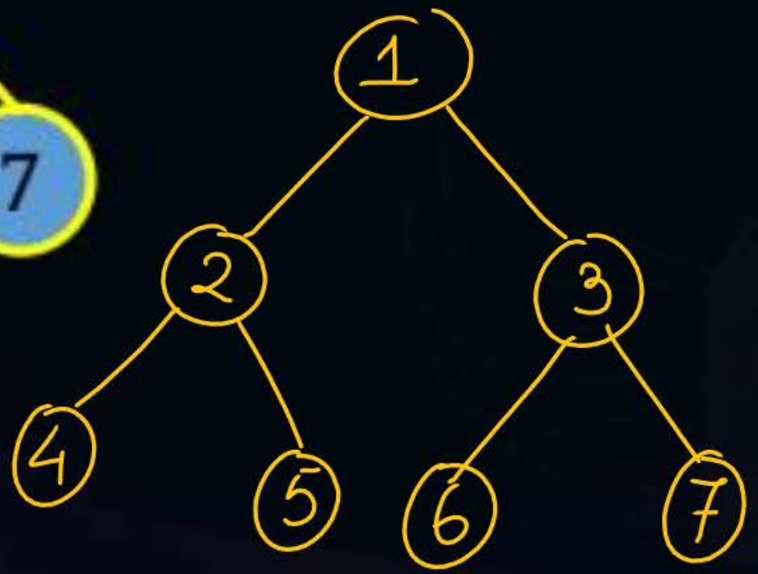
Topic : Graph





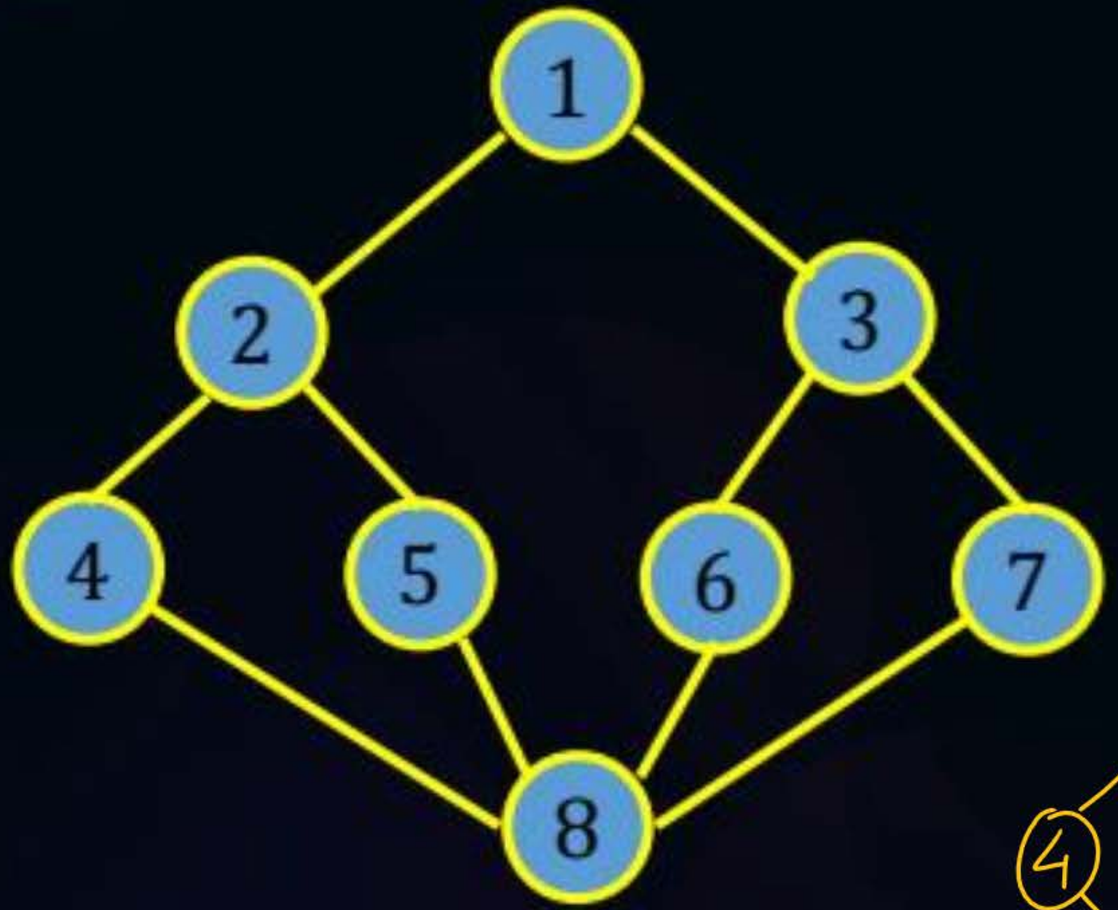


dequeue(7)



Slide

Topic : Graph



Breadth first Search tree



Queue





dequeue(8)

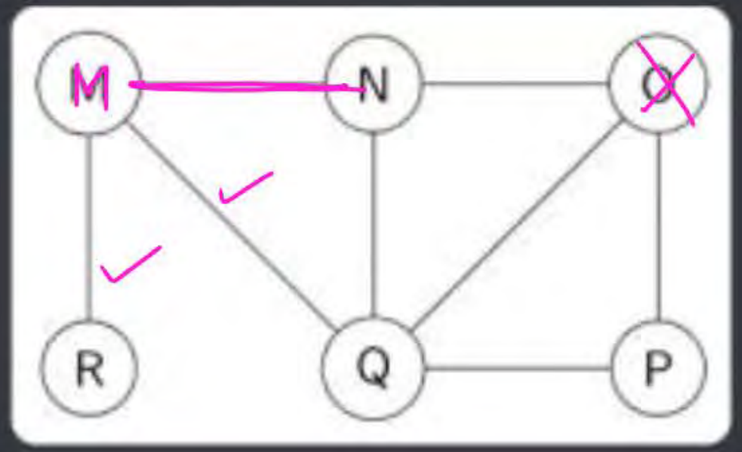
queue Empty

algorithm stops

Slide

Question

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?

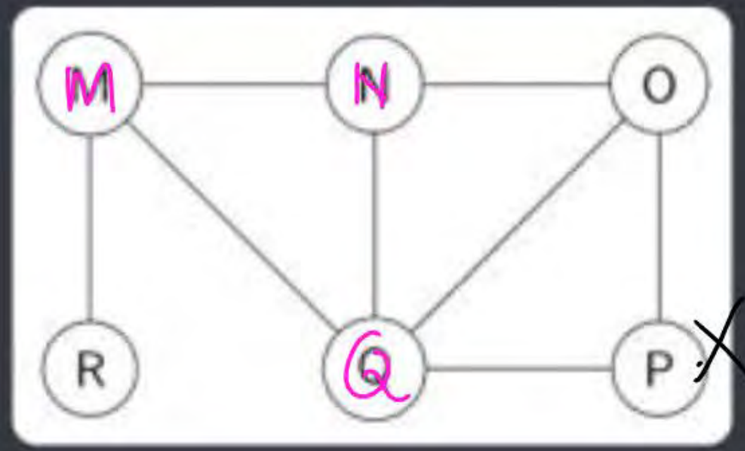


A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

Slide

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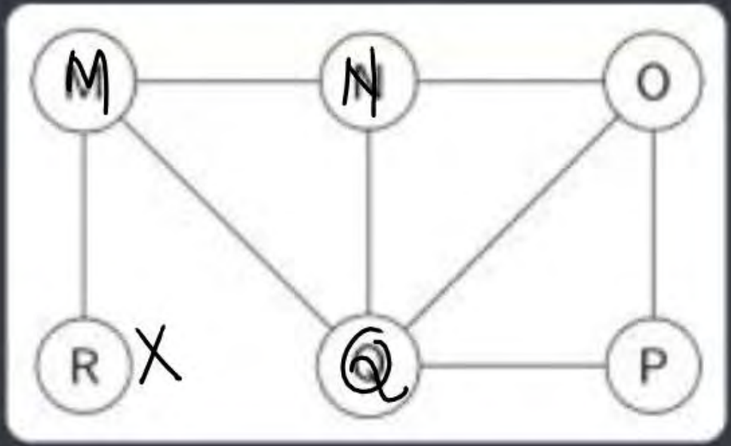


A. MNOPQR
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Slide

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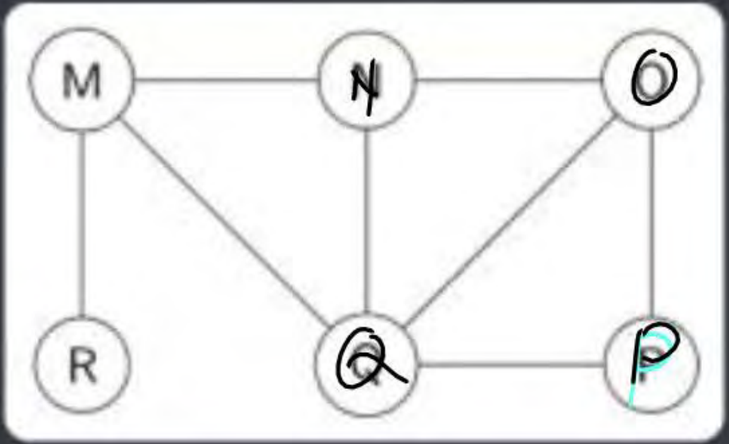


A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

Slide

Question

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

Slide

Question

Consider the following graph:



1, 2, 3, 4, 5

1, 2, 3

1, 3, 2, 7, 6, 5, 4

Identify **valid BFS** search sequence

(a) 1, 3, 2, 5, 4, 7, 6, 8

(b) 1, 2, 3, 6, 7, 4, 5, 8

(c) 1, 2, 3, 4, 6, 7, 5, 8

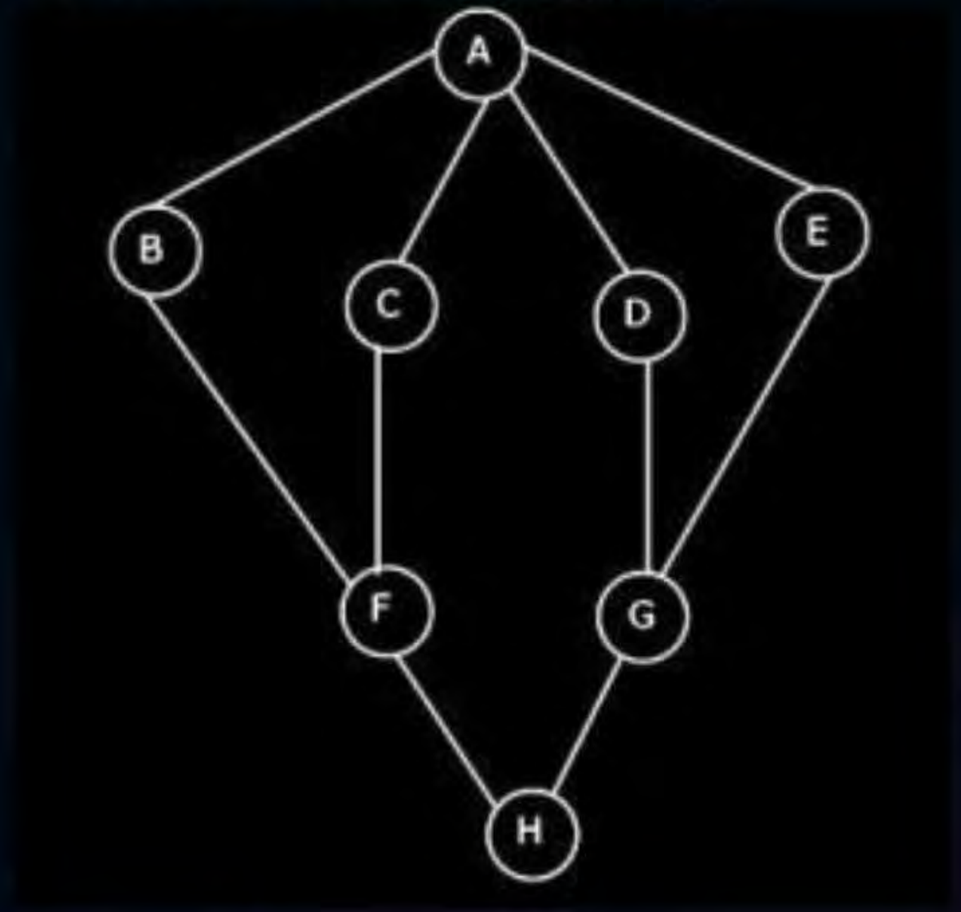
(d) 1, 3, 2, 7, 6, 5, 4, 8

Slide



Topic : Graph

Consider the following graph



How many different breadth-first search traversals are possible considering H as a source vertex?

(A) 1

(B) 4

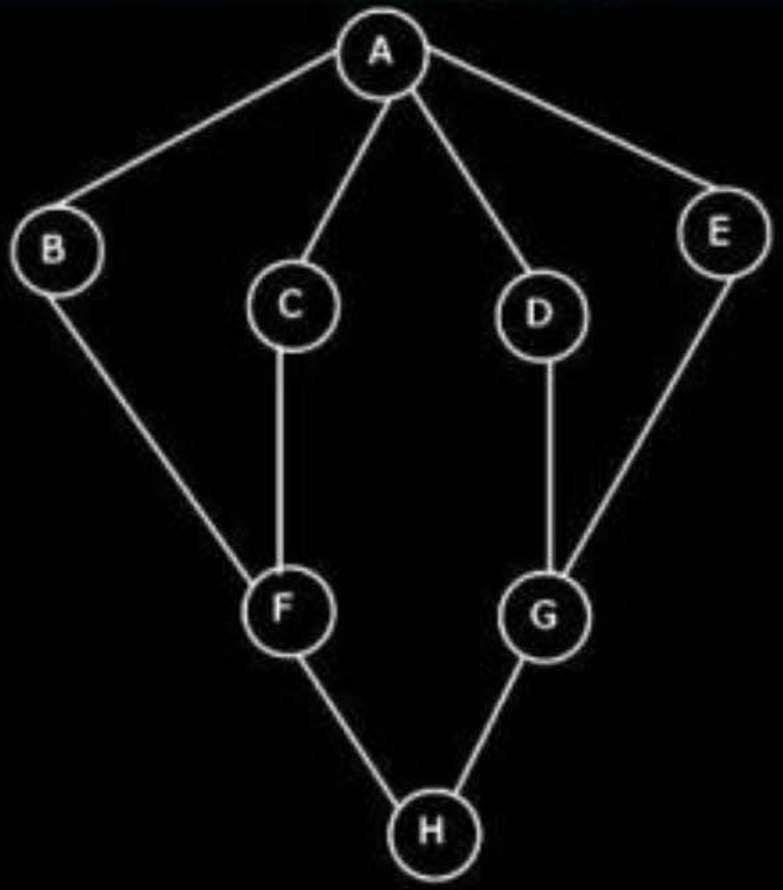
(C) 16

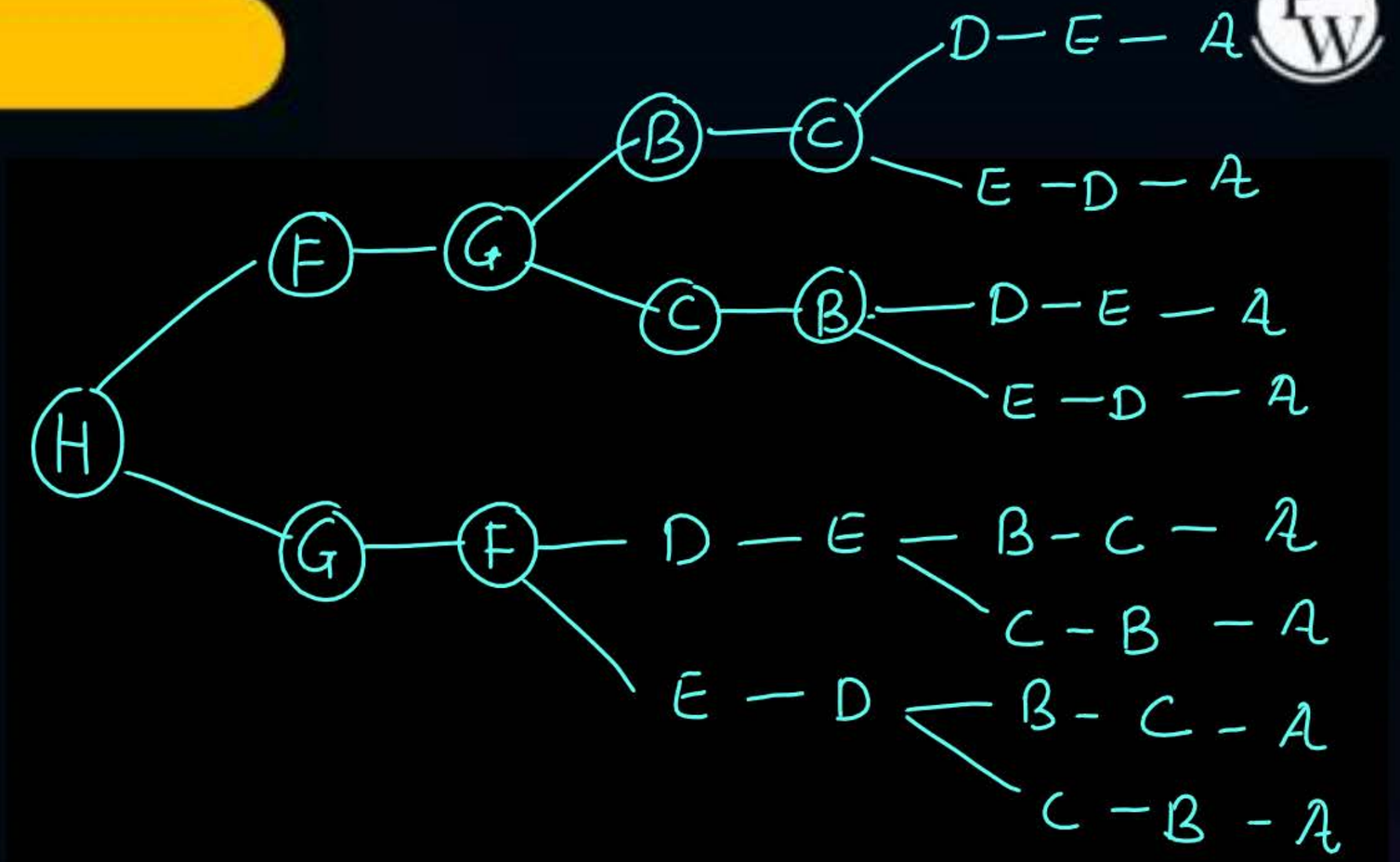
(D) 8

Slide

Topic : Graph

Consider the following graph





Slide



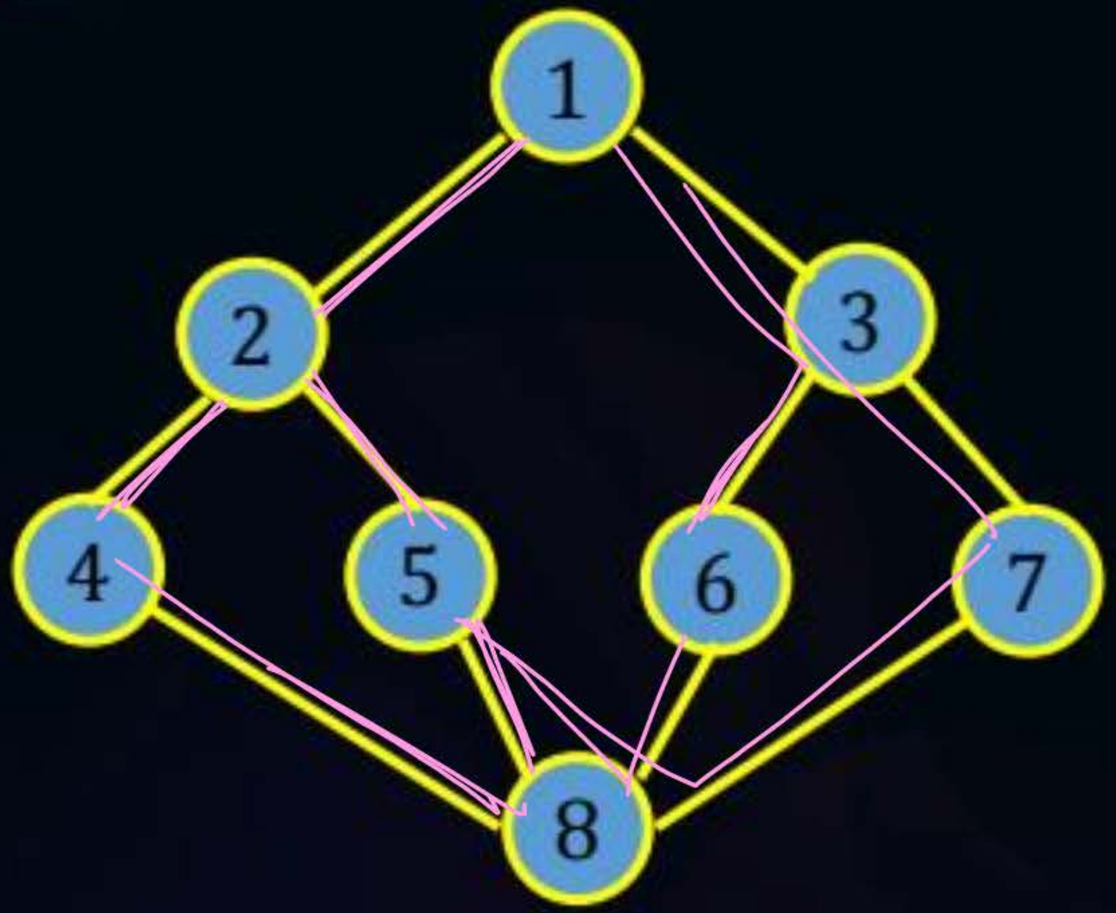
Question

Q. Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the nth vertex in this BFS traversal, then the maximum possible value of n is _____.



Slide

Topic : Graph



① Same vertex 1

Distance of 5 in terms of No of edges

1 - 2 - 5 — distance : 2 ✓

1 - 3 - 7 - 8 - 5 distance 4 ✓

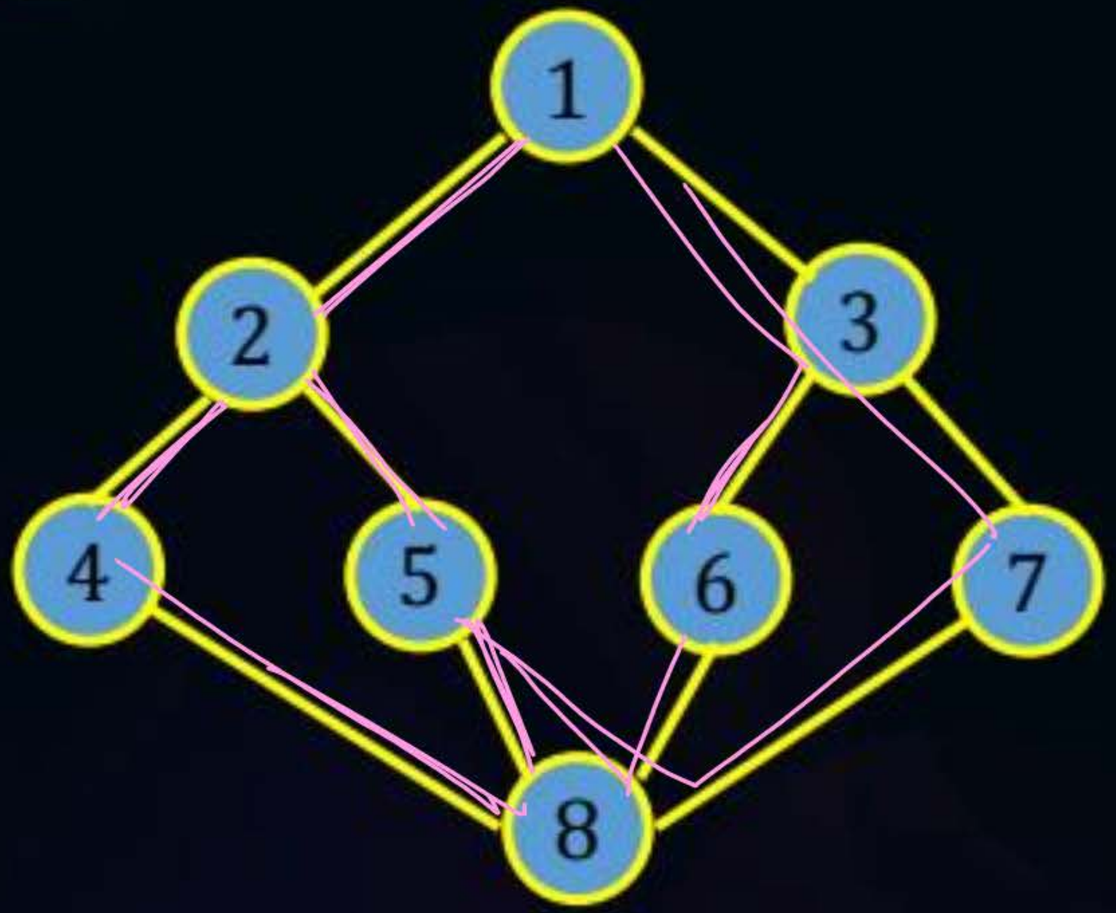
⋮ more

1 - 3 - 7 - 8 - 4 - 2 - 5 — 6 ✓

Slide



Topic : Graph

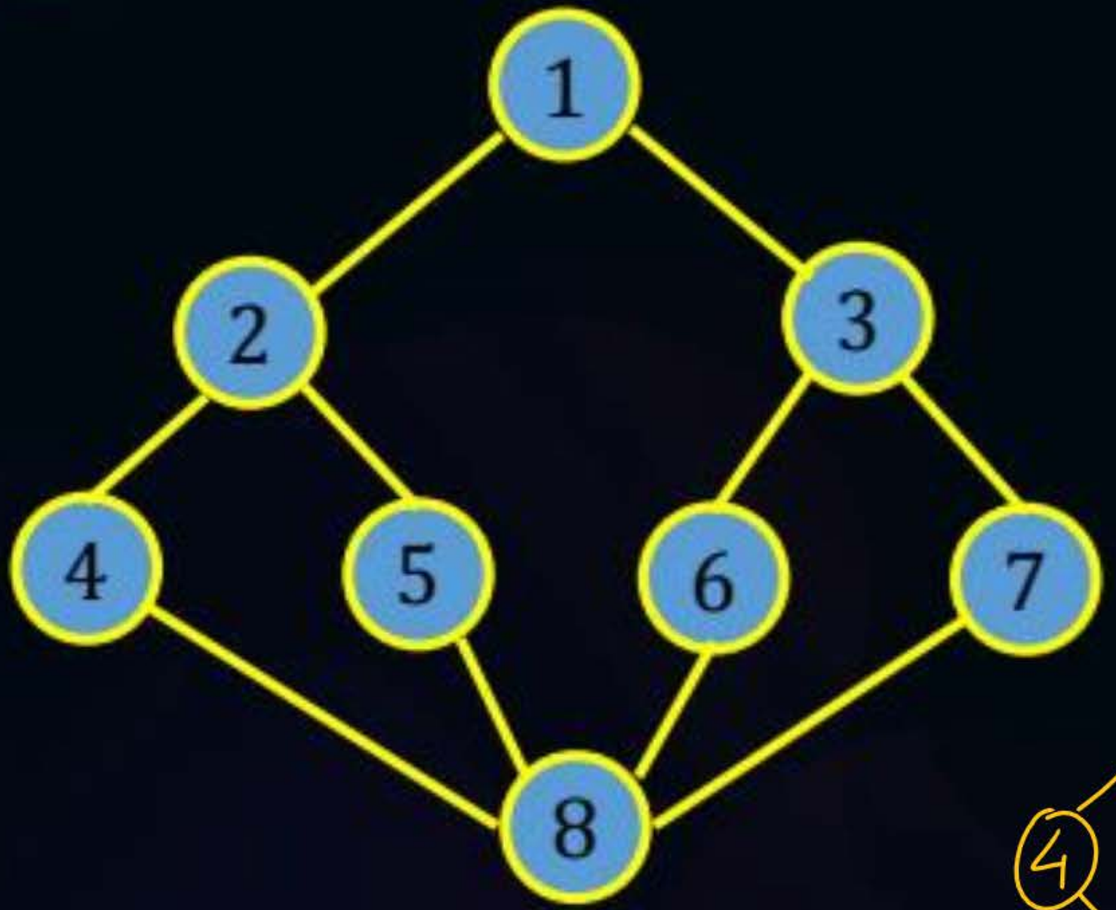


Shortest path algorithm

minimum No. of edges

Slide

Topic : Graph





BFS is a

Shortest path algorithm

for unweighted

undirected graph

Slide



2 mins Summary

Topic

Graph Representation

Topic

Adjacency list

Topic

Adjacency Matrix

Topic

BFS (Breadth first Search)

Topic

Slide

7pm

THANK - YOU

Slide